

Bell Ringer: Algebraic Expressions Introduction



A number m is doubled then decreased by 7.
Write an algebraic expression.

T1

BUILD CONFIDENCE

Define: variable,
constant, expression.

T2

APPLY YOUR SKILLS

A number m is doubled
then decreased by 7.
Write an algebraic expression.

T3

CHALLENGE YOURSELF

The sum of twice a number
and 5 is 15.
Write the equation
(don't solve).



Bell Ringer Answer Reveal

STEP 1

Double the number m .

$$m + m = 2m$$

STEP 2

Decrease by 7
(subtract 7).

$$2m - 7$$

STEP 3

Write the expression
in simplest form.

$$\Rightarrow \boxed{2m - 7} \Leftarrow$$

ANSWER: $2m - 7$



Intermediate Algebra

Unit 1: Lesson 1 –

What Is a Variable?

★ BEGIN THE JOURNEY ★



Explore variables, constants, and expressions—the building blocks of algebra.



MINNESOTA STANDARDS:

9.3.5.3 + 9.3.5.7 + 9.3.5.8



MERIT EDUCATION GROUP



I CAN: What Is a Variable?



A variable is a letter or symbol that represents a number that **can change** or **vary** in an expression.

★ THINK OF IT AS... ★



A placeholder for a number we don't know **yet**.

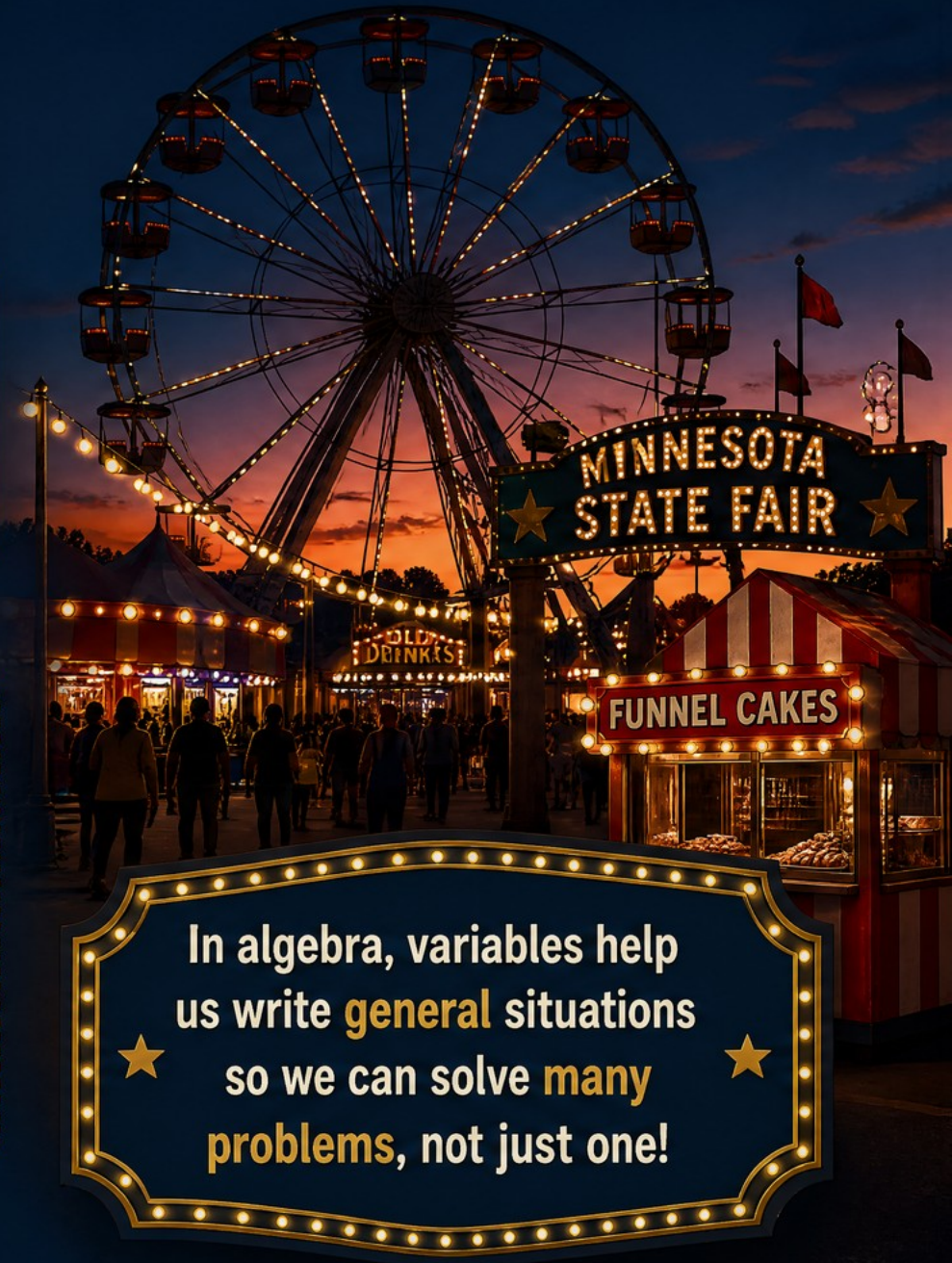


A value that can **change** depending on the situation.



A quantity that can **vary** from one ride to the next.

In algebra, variables help us write **general** situations so we can solve **many** **problems**, not just one!



Vocabulary Cards



VARIABLE

A letter that represents one or more numbers.

EXAMPLE:

x represents any number.



CONSTANT

A number that does not change.

EXAMPLE:

7 is a constant.



EXPRESSION

A combination of numbers, variables, and operations (without an equals sign).

EXAMPLE:

$2m + 5$



EQUATION

A statement that two expressions are equal.

EXAMPLE:

$2m + 5 = 15$



Collaborative THINK-PAIR-SHARE Activity



1. THINK ALONE

Read the prompt.
Think about your approach.
Jot down your ideas.



2. PAIR UP

Partner with a classmate.
Compare your strategies
and your work.



3. SHARE A STRATEGY

Share your partner's or
your strategy with
the class.

★ OUR THINKING FRAME ★

I started by _____ because _____.



$$2x$$

$$x + 5$$

$$3x - 7$$

Worked Example (I DO)

★ TEACHER-LED EXAMPLE ★

Let's translate, write, and simplify!



A number m is doubled
then decreased by 7.
Write an algebraic expression.

TRANSLATION CLUES

A number	→	m
is doubled	→	$\times 2$ or $2()$
increased by	→	$+$
decreased by	→	$-$

STEP 1: Start with the number m .

STEP 2: Double the number.
 $m + m = \underline{2m}$

STEP 3: Decrease by 7.

$$\underline{2m - 7}$$

★ The algebraic expression is $2m - 7$.



Guided Practice (WE DO)

★ We solve problems together! ★



TEACHER SCAFFOLDS

We work through each step together with teacher guidance and support.



YOUR TURN

Answer, ask questions, and share your thinking—we're all in this together!



STEP BY STEP

Follow the plan: understand, plan, solve, check.

Ask • Think
Share • Learn
We grow together!



Your voice matters.



We support each other.



Every step counts!

❖ Ojibwe Word Problem + Culture ❖



An Ojibwe elder counts $4x$ fish in one net and $3x$ fish in another net.

- Write an expression for the total number of fish.
- If $x = 7$ fish per net, how many fish are there total?



READING STRATEGY

CIRCLE



numbers/
operators



UNDERLINE



the question



BOX



operations



WRITE



the expression

OJIBWEMOWIN (OJIBWE LANGUAGE): NUMBERS 1-4

1



bezhig

2



niizh

3



niswi

4



niiwin



FACT: Ojibwe families harvest wild rice (*manoomin*) together from canoes on Minnesota's lakes, measuring and counting each grain with care and gratitude. Manoomin is a sacred food that sustains our people, our culture, and our future.



HONOR.
LEARN.
RESPECT.
TOGETHER.



Ojibwe Math Honor and Culture



MANOOMIN: WILD RICE

Each late summer, Ojibwe families harvest manoomin (wild rice) together from canoes on Minnesota's lakes. We measure and knock the rice into the canoe with aare and gratitude, and a portion of the harvest is set aside for ceremony. This tradition strengthens our connection to the land, our ancestors, and one another.

★ WORKED SOLUTION ★

1

STEP 1

Add the terms:

$$\text{total} = 4x + 3x$$

2

STEP 2

Combine like terms:

$$4x + 3x = 7x$$

3

STEP 3

Substitute $x = 7$:

$$7(7) = 49$$

FINAL ANSWER: 49 FISH

MORE CULTURAL FACTS



Manoomin is a staple food and a gift from the Creator. It provides sustenance for our families and helps us live in balance.



Harvesting manoomin is guided by respect—following the seasons, water levels, and teachings passed down through generations.

OUR COMMUNITY TODAY



Ojibwe people are one of Minnesota's 11 Tribal Nations with deep roots across the state.



We have schools, colleges, and programs that keep our language, culture, and traditions strong.



We contribute to Minnesota every day in health care, education, government, arts, and many other fields.

MINNESOTA
STATE FAIR

Sweet Martha's
COOKIES



Career Spotlight: Medical Lab Scientist



CAREER PROFILE

Medical Lab Scientist



Mayo Clinic

Rochester, Minnesota

Medical Lab Scientists perform tests on samples like blood, urine, and tissue to help doctors diagnose and treat patients.



KEY SKILLS



Interpret formulas to understand test results and lab procedures.



Evaluate expressions by substitution to calculate and verify values in quality control.



ENTRY WAGE

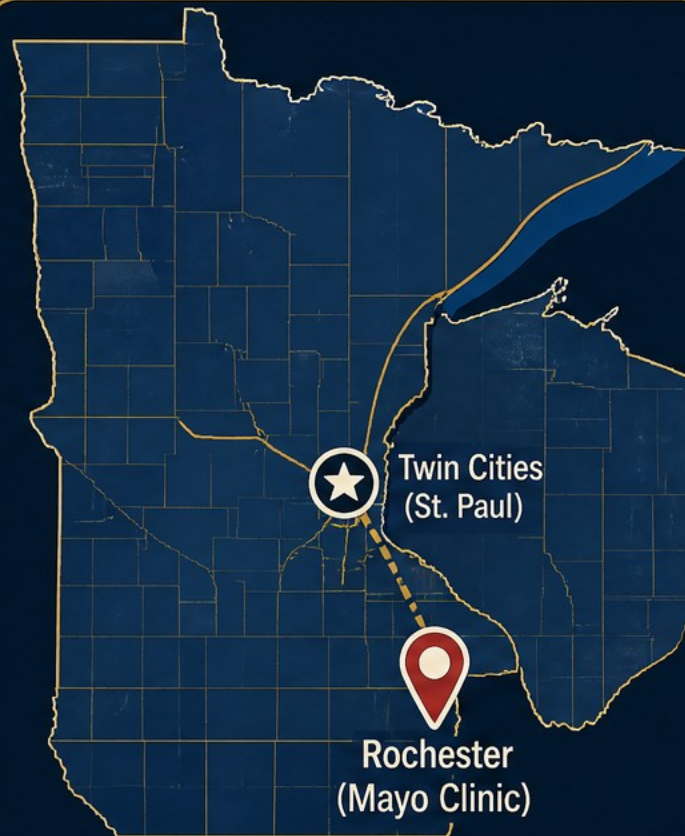
\$63,150 /year

(MN DEED OES data)



EDUCATION REQUIRED

Bachelor's degree in Medical Laboratory Science (or related field) required.



~90 miles south of the Twin Cities



Career Word Problem (Pair Activity)

★ SOLVE LIKE A PRO! ★



REAL-WORLD CHALLENGE: MEDICAL LAB SCIENTIST

A medical lab scientist runs blood tests on samples throughout the day.

- Let n be the number of samples tested.
- The lab uses 5 mL of reagent per sample.
- At the end of the day, 120 mL of reagent remains in the container.



1. Write an algebraic expression for the total amount of reagent (in mL) used.
2. If the scientist tests 25 samples, how much reagent is used?



STRATEGY STATION

THINK LIKE A MEDICAL LAB SCIENTIST!



1. UNDERSTAND

Identify what is changing (varies) and what is known (stays the same).



2. EXPRESS

Write an expression that represents the total reagent used.



3. EVALUATE

Substitute the given value for n and calculate the amount used.



4. CHECK YOUR WORK

Does your answer make sense in the real world?

★ CLEAR THINKING. ACCURATE RESULTS. ★



MERIT EDUCATION GROUP



Communicate your thinking with your partner.
Be ready to share your strategy!



GO Practice: Algebraic Expression

★ PROBLEM ★

A number n is
tripled then
increased by 4.

Write an algebraic expression.



★ GO ANSWER REVEAL ★

STEP
1



Translate the words.

A number n is tripled.
Then it is increased by 4.



tripled → $3n$
increased by 4 → $+ 4$

STEP
2



Write the expression.

Combine the parts in
order.



$3n + 4$

STEP
3



Check your thinking.

Tripling comes first,
then add 4.



$3n + 4$

FINAL ANSWER

$3n + 4$

*We turned words
into an expression!*



Set Practice Problems

1

x decreased by 9

2

The product of 6
and a number y

SWEET TREATS

TICKETS



Set Practice Answer Reveal

★ LET'S REVEAL THE ANSWERS! ★

MINNESOTA
STATE FAIR

1 x decreased by 9.

STEP 1 Start with the number x . x

STEP 2 Decreased by 9 means subtract 9. $x - 9$

STEP 3 Write the expression. $x - 9$

≡ **ANSWER:** $x - 9$ ≡

2 The product of 6 and a number y .

STEP 1 Start with the number y . y

STEP 2 The product of 6 and y means $6 \times y$. $6 \times y$

STEP 3 Write the expression. $6y$

ANSWER: $6y$ ≡

FUN
FOOD
MEMORIES

SWEET
TREATS

Extra Practice Problem



**Twice a number
increased by 3
equals 15.**

**Write an equation and
find the number.**



MERIT EDUCATION GROUP



Extra Practice Answer Reveal

STEP 1

Write the equation.

$$2n + 3 = 15$$

STEP 2

Subtract 3 from both sides.

$$\begin{array}{r} 2n + 3 = 15 \\ -3 \quad -3 \\ \hline 2n = 12 \end{array}$$

STEP 3

Divide both sides by 2.

$$\begin{array}{r} 2n = 12 \\ 2 \quad 2 \\ \hline n = 6 \end{array}$$

FINAL ANSWER

$$2n + 3 = 15$$

$$n = 6$$

MINNESOTA
STATE FAIR



Merit Time Game: Translation Match



SETUP

- This is a card/slide-driven game.
No physical board!
- Cards (slides) show phrases or situations.
- Match each with its algebraic expression.
- Your device = your game space.



PLAY

- One card (phrase/situation) appears.
- Choose the card with the matching algebraic expression.
- Think first, then select your match.
- You'll see the answer and explanation before the next round.
- Work at a steady pace and try to build your streak!



SCORING

- +10 points for each correct match.
- Keep your streak going for bonus points!
- Final score = base points + Merit bonuses.



★ PBIS EXPECTATIONS (Game Master Rules) ★



READY

Devices ready.
Focus your thinking.
Be ready to play!



RESPONSIBLE

Answer on your own first.
Make thoughtful matches.
Stay on task and keep going!



RESPECTFUL

Use read words.
Listen to others.
Celebrate everyone's success!



EARN MERIT BONUSES!



VELOCITY

Finish rounds quickly and accurately!
Up to +30 pts



READINESS

Be ready to start right away.
Up to +20 pts



STREAK

Keep your streak of correct matches!
Up to +30 pts



Individual Solve: Tiered Problems



T1 BUILD

Build Confidence

Define: variable,
constant, expression.



T2 APPLY

Apply Your Skills

A number m is doubled
then decreased by 7.
Write an algebraic
expression.



T3 CHALLENGE

Challenge Yourself

The sum of twice a
number and 5 is 15.
Write the equation
(don't solve).



ACCESS VIA: PROBLEMS: TIERED WORKSHEET - <LESSON> - IN YOUR LMS



SHOW YOUR WORK



MERIT EDUCATION GROUP

LEARN
SOLVE
SUCCEED



MINNESOTA
STATE FAIR

FUN
FOOD
FRIENDS



Exit Ticket



**SHOW YOUR WORK
FOR FULL CREDIT**



SUBMIT YOUR WORK IN YOUR LMS

- ★ Upload your completed work.
- ★ Make sure your response is submitted before you leave class.



MERIT EDUCATION GROUP



Recap and What's Next

WHAT WE DID



Understood Variables

We learned what a variable is and how it represents a number.



Built Expressions

We wrote algebraic expressions using variables, numbers, and operations.



Applied and Evaluated

We evaluated expressions by substitution and solved real-world problems together.

WHAT'S NEXT



We'll simplify and combine expressions in the next lesson.

